Calculate: if VLM P99 = 1500ms, state bridge overhead = 2ms, IK command execution = 10ms — what is the effective planning rate?

Step 1: Given Parameters (Worst-Case Pipeline)

- VLM Tail Latency ($P99$): $\tau_{\text{VLM}} = 1500$ ms
- State Bridge Overhead: $\tau_{\text{bridge}} = 2$ ms
- IK Command Execution Time: $\tau_{\text{IK}} = 10$ ms

Step 2: The Core Calculations

1. Maximum Safe IK Loop Frequency ($f_{\text{IK\_max}}$)

As established in Stage 1, the local hardware-actuation loop depends strictly on the local communication and processing times:

$$\tau_{\text{ctrl}} = \tau_{\text{bridge}} + \tau_{\text{IK}} = 2\text{ ms} + 10\text{ ms} = 12\text{ ms}$$

$$f_{\text{IK\_max}} = \frac{1}{\tau_{\text{ctrl}}} = \frac{1}{0.012\text{ s}} \approx 83.33\text{ Hz}$$

2. Total Worst-Case System Latency (τ_{total})

Because the system runs asynchronously, a complete perception-to-actuation planning cycle relies on the full sequence of vision inference, data marshaling across the bridge, and immediate motor execution:

$$\tau_{\text{total}} = \tau_{\text{VLM}} + \tau_{\text{bridge}} + \tau_{\text{IK}}$$

$$\tau_{\text{total}} = 1500\text{ ms} + 2\text{ ms} + 10\text{ ms} = \mathbf{1512}\text{ ms } (\mathbf{1.512}\text{ s})$$

3. Effective Planning Rate (f_{planning})

The **Effective Planning Rate** is the absolute frequency at which the robot receives fresh, updated cognitive path waypoints from the closed-loop system under worst-case network conditions. It is determined by the total system latency:

$$f_{\text{planning}} = \frac{1}{\tau_{\text{total}}} = \frac{1}{1.512\text{ s}} \approx \mathbf{0.661}\text{ Hz}$$

Step 3: Architectural Interpretation for the Paper

Under worst-case (*P99*) conditions:

- The local arm moves smoothly and safely at 83.33 Hz (updating its joint trajectory commands every 12 ms
- The cognitive VLM visual planner drops an updated target waypoint into the buffer once every 1512 ms resulting in an effective planning refresh rate of 0.66 Hz